



2025 AI in Education

A Microsoft Special Report



Introduction

AI is transforming nearly every aspect of our society, from how work gets done to how companies operate. It's changing how we talk with our doctors, our employees, and our customers, bridging language gaps and helping us communicate more clearly and effectively. And it's transforming education too.

Across the globe, educators are using AI to increase student agency, giving them a greater sense of ownership over how they learn. Education leaders and institutions are leveraging AI to enhance operational and administrative processes and are tapping into real-time data to improve student success. And an exciting shift is starting when it comes to the benefits of using AI—no longer just a time saver, it can empower everyone to reimagine opportunities for the future of education together.

To better understand the state of AI in education, Microsoft Education has conducted numerous studies and surveys and collaborated with academic institutions and organizations. We've also surveyed academic and IT leaders, educators, and students from around the world; explore the extended survey data [for more insights](#).

A follow-up to our [AI in Education Report in 2024](#), this year's edition reveals much about the ongoing evolution of AI in education, from how audiences feel about it to how it's being used in education today—and how it might be used in the future.

The Age of Creation and Conversation

Over the past year, AI has emerged as a creative thought partner in the classroom. Leaders at Fulton County Schools in Georgia wanted to create more personalized learning experiences for their 87,000 students across 104 schools. After training educators and students on how to use the tool and create a structured and protective environment, the district wove Copilot Chat into the curriculum. They quickly saw an increase in student confidence and curiosity. “We use Copilot Chat as a brainstorming partner to ideate, but not to actually do our work for us,” said Pragya Modgil, a junior in the district. “It helps us collaborate and expand our creativity to think of more ambitious ideas.”

“I see great examples where AI is used, not just in a one-to-one situation—one kid in front of a computer—but a group or a whole class using it as a catalyst for conversation. This is the age of conversation. It’s fueled by AI, but it’s about the power of conversation and dialoguing, and that’s a very human experience.”

Mark Sparvell,
Director, Marketing Education,
Microsoft

1

Who's Using What, Where

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There is no question that AI adoption in education is widespread: 86% of education organizations in an IDC study now report that they use generative AI, the highest rate for any industry.¹ The latest survey from Microsoft Education reflects similar results.



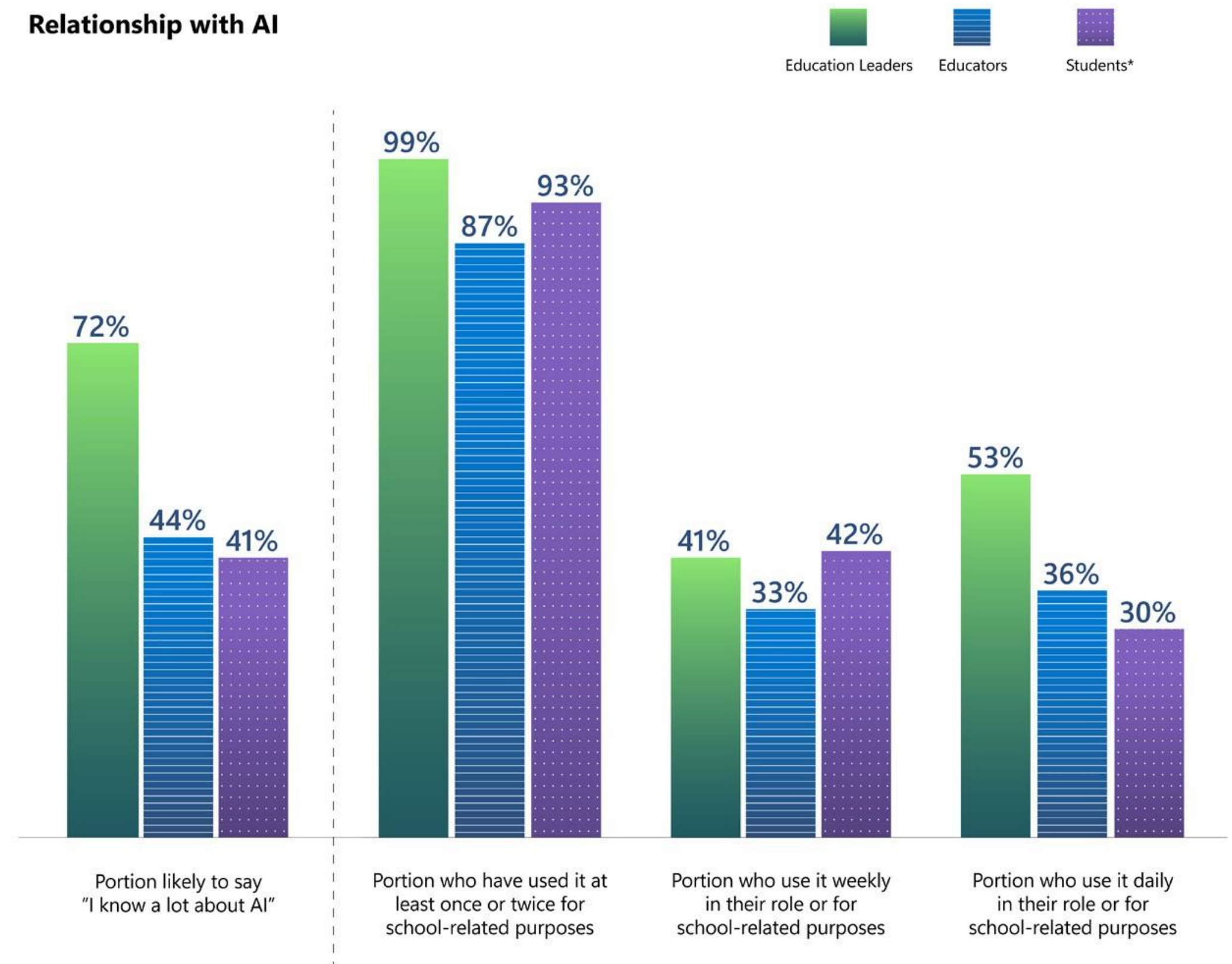
A Steep Adoption Curve

- The percentage of students and educators in the US¹ who say they use AI “often” for school-related purposes jumped from last year by 26 points and 21 points, respectively.
- The portion of US students who say they have never used AI dropped by 20 points from last year.
- Notably, while most respondents in all groups have used AI at least once (with many who use it daily or weekly), less than half of educators and students said they knew a lot about it, a disconnect that points to the importance of AI discussions, training, and literacy.
- Discover more insights in the [extended survey data](#).

Usage Is High, Around the World

Education leaders are the most active AI users, but students and educators aren't far behind.

Relationship with AI



*Student figures are US only

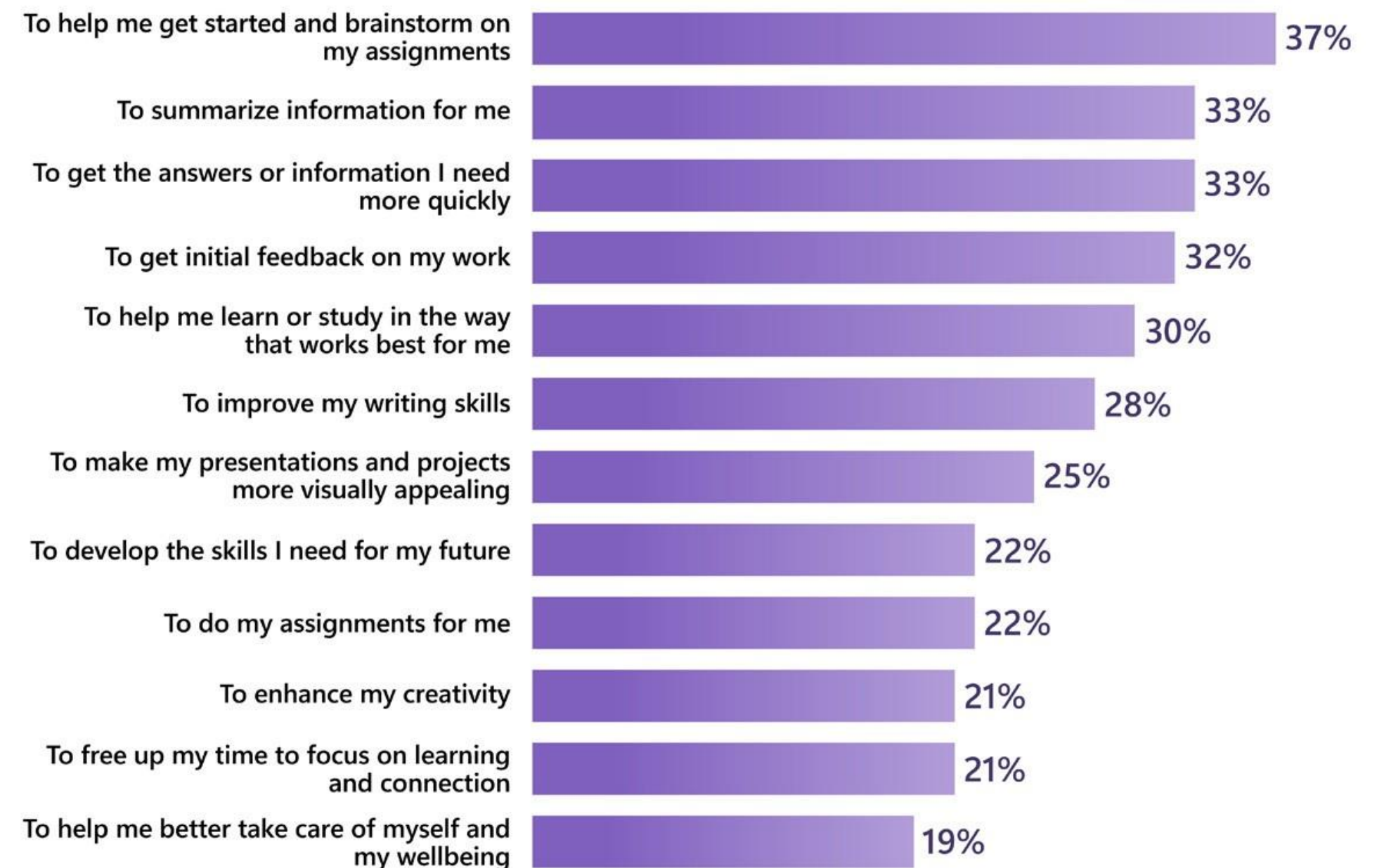
AI for Students

Students' familiarity, use, and comfort with AI has increased dramatically; they often use it to save time, improve work, and assist with learning.

In a recent study from Australia, university students who used an AI-powered chatbot saw a nearly 10% improvement on their exam grades over peers who weren't using the tech. Predictably, AI use peaked during the weekend before the final exam; after the test, 72% of users stated they would be very disappointed if they couldn't use it again.

Student Use Cases

From brainstorming to receiving feedback, students turn to AI as a conversation partner to help them learn and study in ways that work best for their situation.



Students in the US showed an 11-point jump over last year in three categories of AI usage: using it to develop the skills they need for the future; to learn and study in the way that works best for them; and to help take better care of themselves and their wellbeing.

AI for Educators

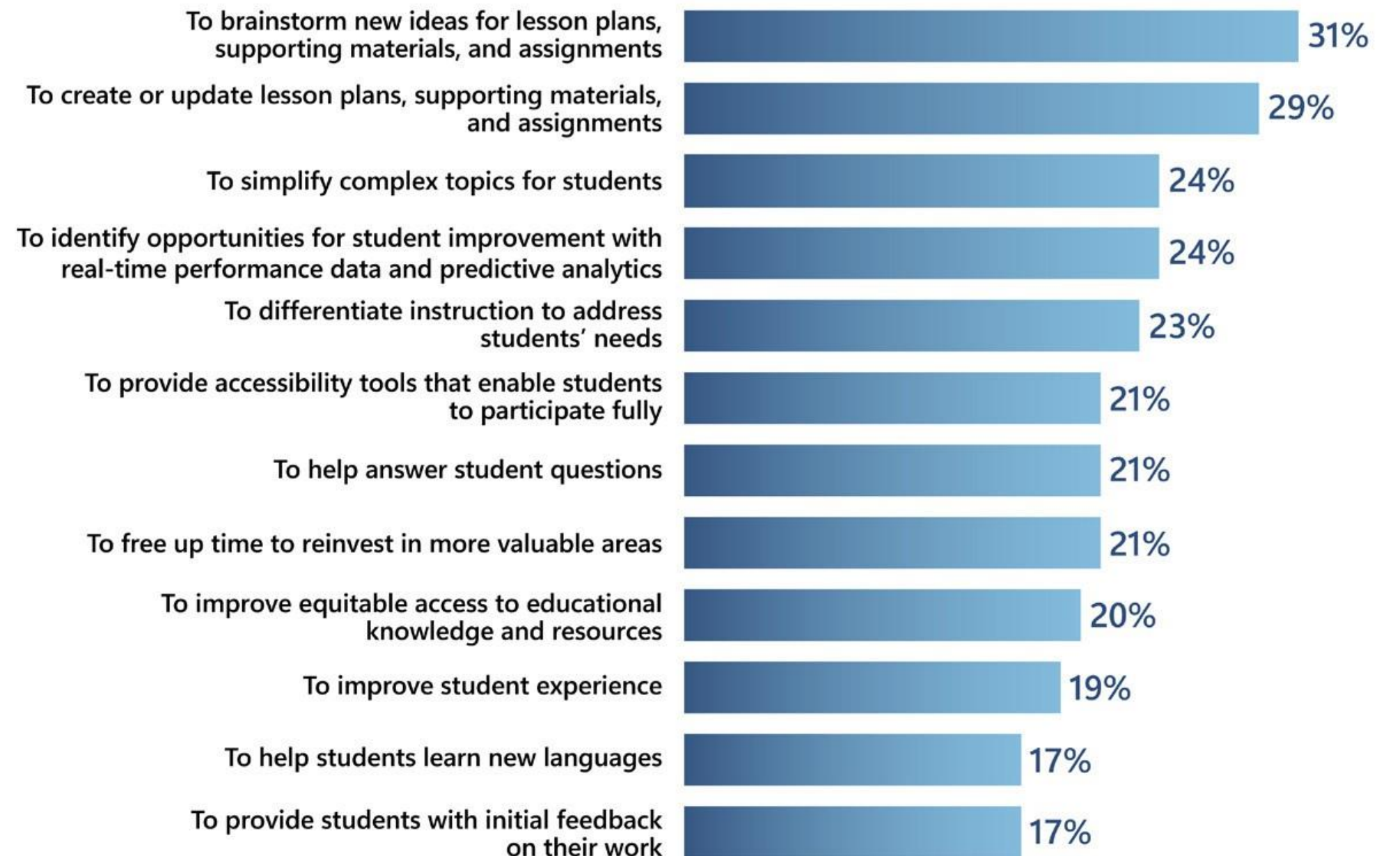
Creating materials for class and answering common questions from students are time-consuming jobs for educators, who increasingly leverage AI to support these core tasks. Many have also expanded their use of AI to help students learn inside and outside of the classroom.

The [Education Authority of Northern Ireland](#), which encompasses 21,000 educators and 380,000 students, integrated Microsoft 365 Copilot to reduce teacher workload, enabling them to more quickly prepare lesson materials like PowerPoint presentations and develop resource materials that can address diverse learning styles.

Faculty at the [University of Manchester](#) are using Microsoft 365 Copilot to support curriculum development and design, expedite research processes, tailor learning materials, and save time. They're also focusing on sharing knowledge, building AI skills, and exploring the potential of agents.

Educator Use Cases

Educators use AI to create classroom materials and personalize instruction.



K-12 educators outside the US were 8 points more likely than those in the US to say they used AI to improve equitable access to educational knowledge and resources.

AI for Education Leaders

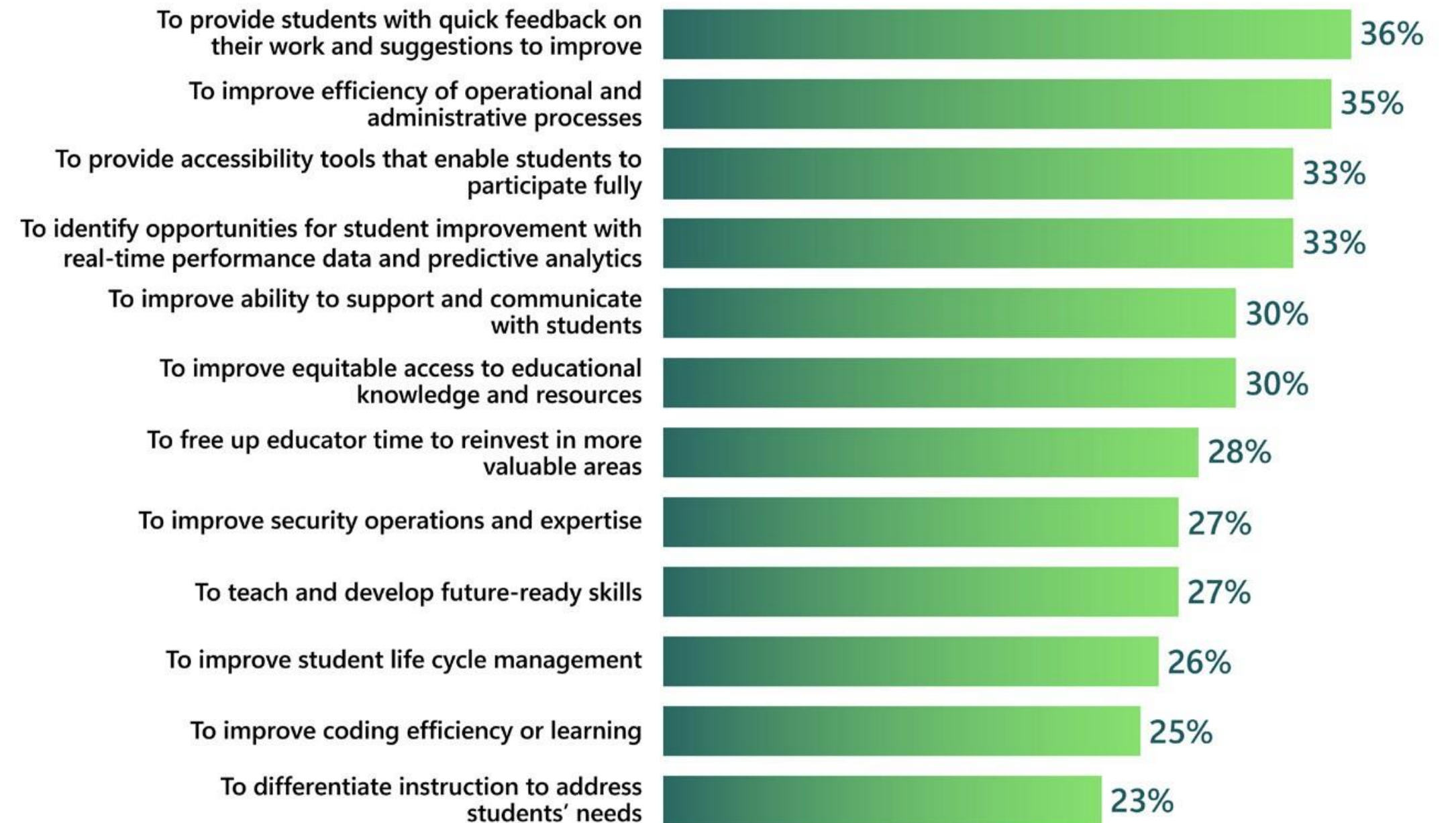
AI can help IT and academic leaders with administrative tasks and operational efficiencies. It can also help to improve communications and enable more students to fully participate.

Answering questions takes time and patience. At [Brevard Public Schools in Florida](#), the IT team created an AI-powered chatbot to staff their help desk, answering queries from students and parents throughout the district.

AI can help bridge language barriers in schools with diverse populations. At a [primary school in Czechia](#), administrative staff use AI-powered translation tools to communicate better with students and parents, 15% of whom come from abroad, including Afghanistan, Ukraine, and Mongolia.

Leader Use Cases

Leaders use AI to improve administration and operations, as well as to improve student performance with real-time data and predictive analytics.



Year-over-year changes in preferred use cases vary considerably between US K-12 IT leaders, higher education IT leaders, and academic leaders. Review the [survey data details](#).

From a Boost in the Classroom to a Boost in the Workplace

According to our survey, 33% of leaders use AI to provide accessibility tools that help students participate more fully, a benefit that is also borne out in workplace studies: a recent study from [EY](#) and [Microsoft](#) found that more than three-quarters of self-reported neurodivergent participants said that Copilot helped them perform better at work, improved the quality of their work, and boosted feelings of inclusion.

It helped 80% of them with written communication, while 90% reported that it made key tasks, like understanding meeting contributions and creating follow-up actions, more accessible and efficient.

An Accessibility Assist

Copilot has helped neurodivergent and disabled employees improve their work experience.



85%

report that Copilot helps them **perform better in their role**



83%

confirm it helps them **improve the quality of their work**



77%

say that it can help **bring out the best in them**

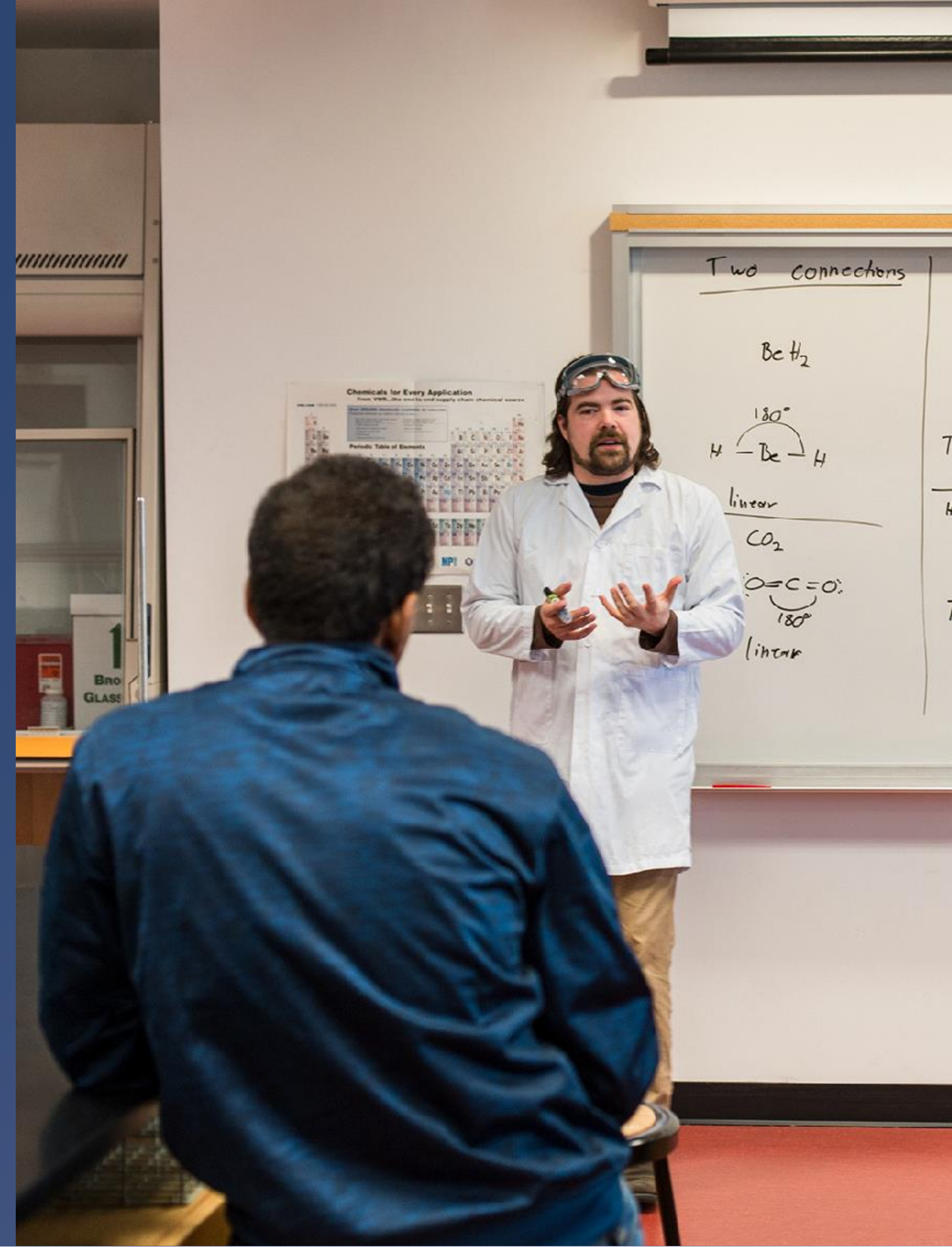


85%

see Copilot as **supportive of their inclusion** at work

A Complement to Traditional Learning

AI can provide opportunities to deepen learning and make it more enjoyable for students. In a study conducted by Microsoft Research and Cambridge University Press & Assessment, students preferred an AI reading assistant over traditional note-taking, citing its ability to answer questions, simplify complex material, and provide immediate feedback. Most students asked sophisticated questions of their AI chatbots to gain a deeper understanding of the material. However, using AI alone produced worse results than using it in tandem with traditional methods.



A Varied Impact

Not all students equally benefit from AI. Students in higher-ranking universities who have greater familiarity with AI experience greater benefits, according to the recent study of university students from diverse backgrounds and usage patterns. Students using AI did better on exams but notably less well than they did on assignments, suggesting that the benefits of using AI to complete assignments didn't transfer to autonomous test situations. One promising find: socioeconomic status had no significant effect on outcomes, suggesting that AI-assisted education does not worsen existing social inequalities.¹

In Nigeria, World Bank conducted a randomized controlled trial using Microsoft Copilot to enhance English language learning for first-year senior secondary students. According to their paper, "the intervention demonstrated a significant improvement of 0.31 standard deviation on an assessment that included English topics aligned with the Nigerian curriculum, knowledge of artificial intelligence, and digital skills." Dive into [the report](#) to review their full findings.



Recommendations

- 1 **Engage with educators and students** to find out what's working well and where there are additional opportunities.
- 2 **Embrace experimentation** with new ways of enhancing learning by using AI to complement traditional learning methods—not to replace them.
- 3 **Invite students to the table** to provide input on institutional AI plans.



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Getting on the Same Page About AI Usage

Getting on the Same Page About AI Usage

While familiarity and usage are high across all groups, concerns still remain, and barriers like literacy gaps, differing perceptions about how and when AI should be used, and lack of training highlight the need for support.

It will be critical for leaders, educators, and students to work together to bridge divides, learn in tandem, and adapt in real time.



The State of AI Integration

Leaders and educators differ in how they view the current state of AI integration and literacy, but students¹ are clear that knowing how to use AI is essential.

Global

Is AI currently integrated into your school's or district's curriculum?



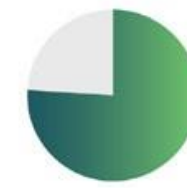
82% of leaders agree



54% of educators agree

Global

I view AI literacy as an essential component of basic education for every student.



76% of leaders agree



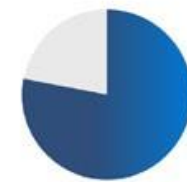
54% of educators agree

Global

I feel confident in my ability to use AI effectively and responsibly.



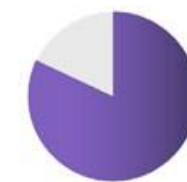
95% of leaders agree



78% of educators agree

US

Knowing how to use AI effectively and responsibly is important for my future.



82% of students agree

¹ US students only; global student data not available.

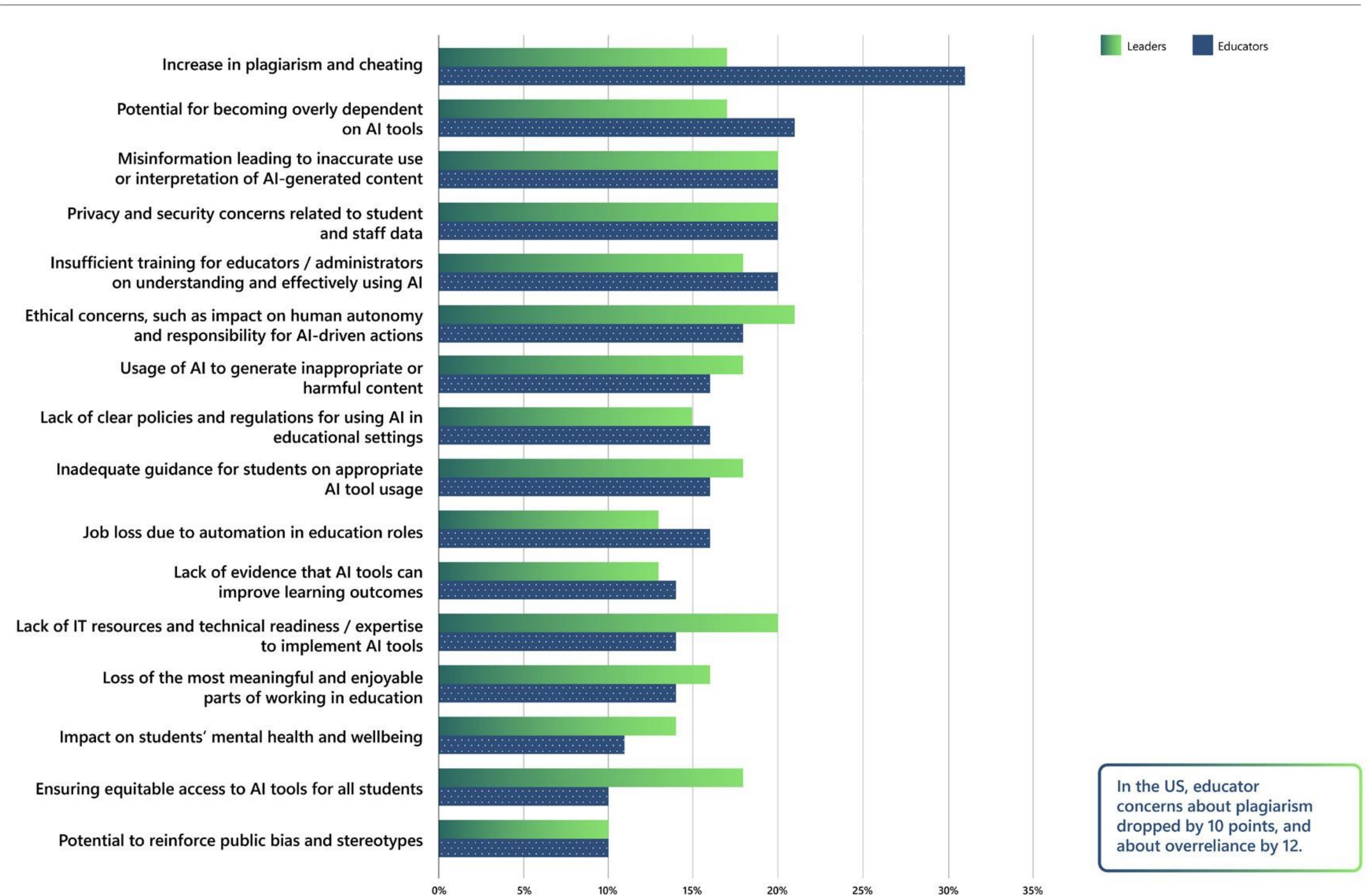
Concerns and Considerations

Concerns over plagiarism and cheating have decreased since last year, but they continue to be top of mind for educators. For students, it is the fear of being accused of plagiarism that worries them. Education leaders are concerned about privacy and security, misinformation, and lack of IT resources or readiness to implement AI tools.



Educator and Leader Concerns

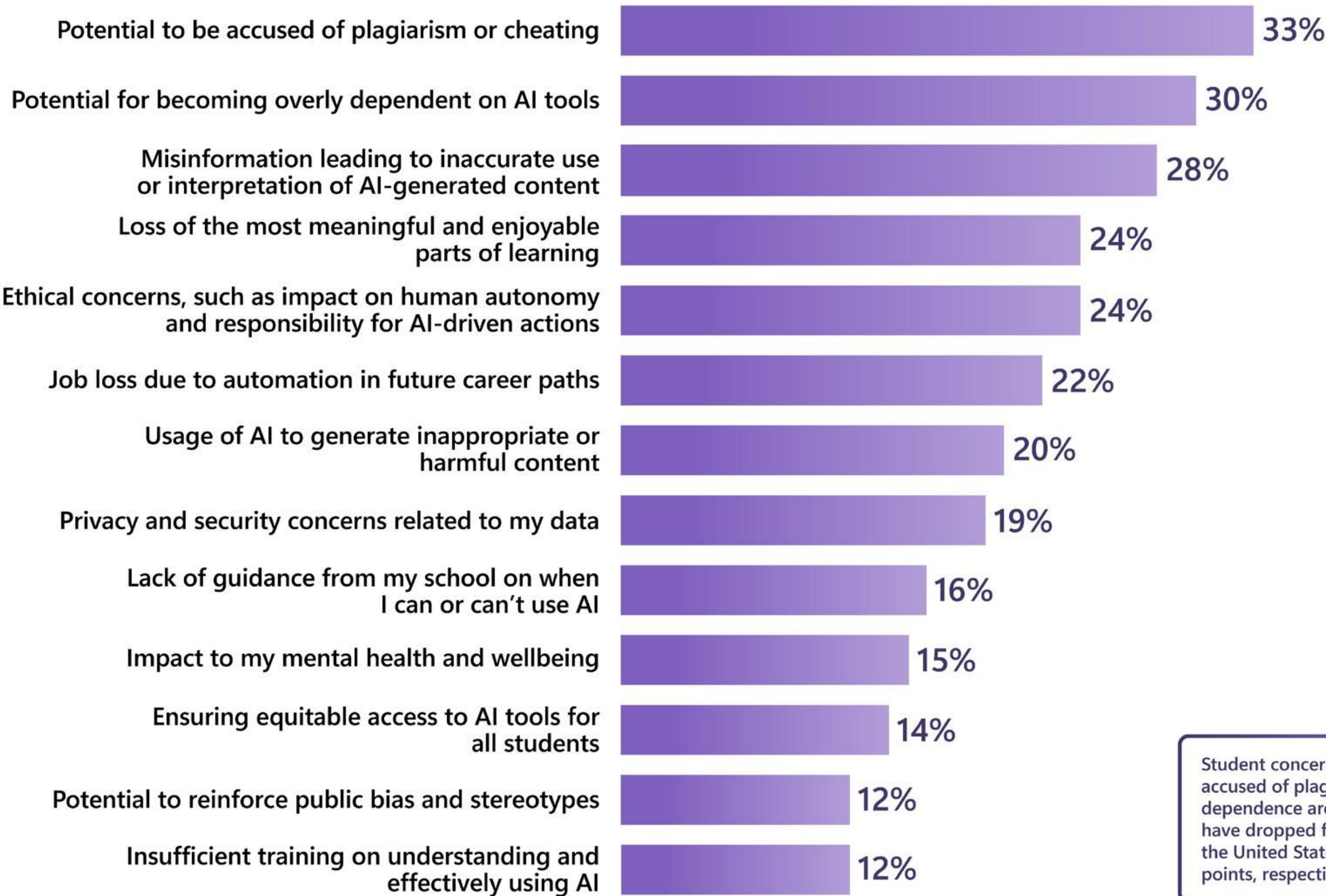
Educators are significantly more concerned about plagiarism than anything else, with additional concerns regarding overreliance, misinformation, security, and insufficient training. For leaders, ethical concerns, lack of resources, and ensuring equitable access also rise to the top.



Survey Question: Please select the top 3 reasons why you may be concerned about AI usage within your organization.
Source: AI in Education Microsoft Study, 2025

Student Concerns

Students are concerned about being accused of plagiarism and becoming overly dependent on AI. Additional concerns include loss of the most meaningful and enjoyable parts of learning, job automation, and inappropriate uses of AI.



Student concerns about being accused of plagiarism and over dependence are high, but they have dropped from last year in the United States, by 19 and 18 points, respectively.

Increased AI Literacy and Training Needs

Students and educators are still gaining the AI knowledge and skills they need as new standards are being defined, such as the [AI Literacy Framework](#) from the European Commission, OECD, and Code.org. Given the speed at which AI is improving, there is a substantial learning curve and, in many schools and districts, an urgent need to increase training opportunities.

Internationally, 76% of academic and IT leaders say that half or more of AI users at their institution had received training, but 45% of educators globally and 52% of students in the US said they had not received any training. This gap points to both a need for more training and a need for leaders to focus on the gap between what they feel they have provided and what educators and students feel they have received. (Explore the [survey data](#) details.)

“Teachers are saying, ‘I need training, it needs to be high quality, relevant, and job-embedded.’ I hope that administrators will not say ‘no one’s allowed to use AI’ or ‘AI’s the future, figure it out.’ In reality, people require guidance, and that means teachers *and* administrators going through professional development.”

Pat Yongpradit,
Chief Academic Officer,
Code.org / TeachAI

Working Together to Address Challenges

Now more than ever, it's important for educators and students to collaborate on challenges.

A key concern is privacy and security—and with good reason: according to a recent Microsoft Digital Defense report, Education and Research was the second-most-targeted sector (21%) by nation-state threat actors in 2024. Some students have joined the fight. At universities across the country, AI is being used to battle cybercrime, and students assist their school's security operations centers (SOCs) and use Security Copilot. It's a win-win: universities get the IT resources they need, while students get on-the-job cybersecurity training.

In the Kent School District in Washington State, teachers use Minecraft Education to bridge the digital divide, building skills like AI literacy and coding among their diverse student body by harnessing the lure of the popular video game. For educators lacking in AI literacy themselves, the district offers intensive content-based training tailored to their specific subjects.

Another creative approach to training: At the University of Sydney, professors build their own trusted AI "agents" to act as virtual TAs. The professors are empowered by becoming an active part of the AI process while modeling appropriate usage for their students.

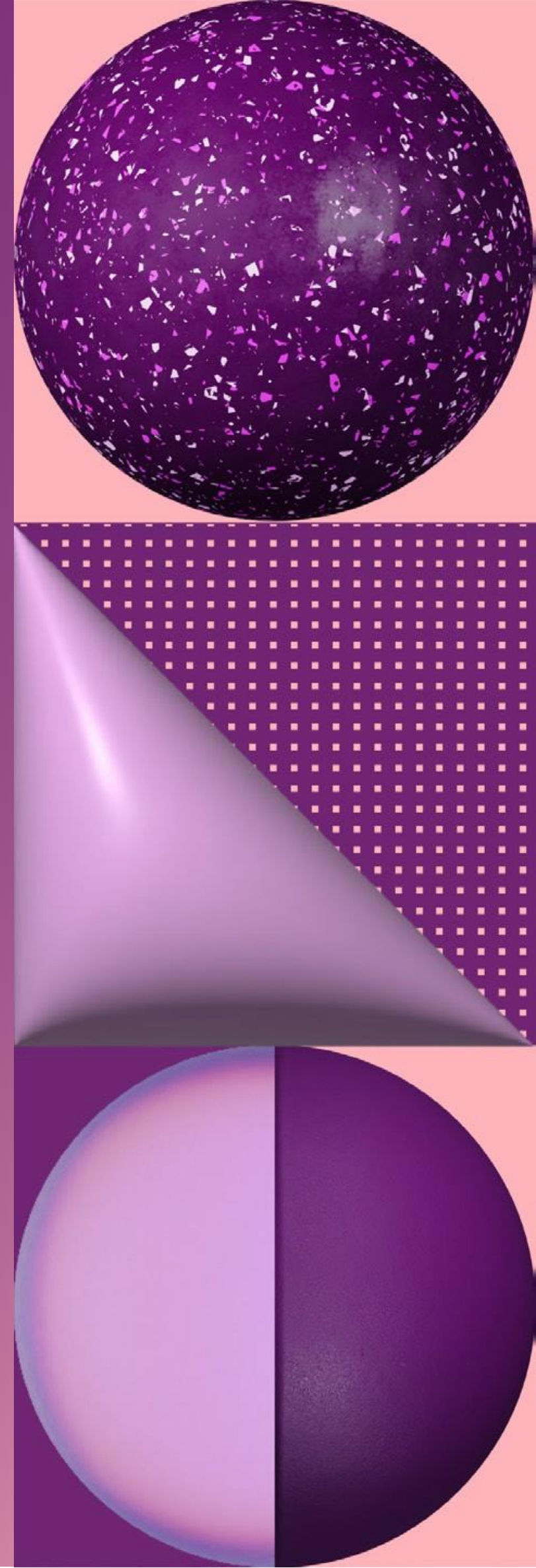
Recommendations

- 1 **Foster open communication**, develop clear guidelines, and solicit feedback from all audiences.
- 2 **Lean into always-on training opportunities** and collaborative, iterative approaches.
- 3 **Create space for your community** to openly share and discuss AI concerns, and work toward solutions together.



3

Essential Skills for the AI Era



Essential Skills for the AI Era

AI fluency is the most in-demand skill for new workers in the US—alongside more human skills like conflict mitigation and adaptability. Those skills, working in tandem, are going to be vital for students entering the new world of work. As a recent [AI and the Global Economy](#) report from LinkedIn noted: “Preparing the workforce with the skills needed to thrive in an AI future is not just an opportunity—it is an imperative.”



Increasing AI Proficiency

Microsoft's latest Work Trend Index Annual Report explored how AI is transforming today's workplace and the skills needed to thrive in these Frontier Firms, where intelligence is available "on tap." Data from that research showed that 71% of education leaders say their institution is considering adding AI-focused roles, and 75% are confident they'll use agents to expand workforce capacity in the next 12 to 18 months.

When it comes to the workforce of the future, upwards of 47% of leaders consider upskilling employees in AI as the top workforce strategy for the next 12 to 18 months, while 78% are considering hiring for AI-specific roles to prepare for the future. Key AI skills that today's students (and thus, tomorrow's workers) will need, according to the report:

- Knowing how to use AI as a colleague and team member, not as a tool
- Learning how and when to delegate to AI, versus human workers
- Thinking like a manager—since everyone will be managing AI

"My theory is that educational institutions are going to start to need to think about, how do we essentially produce early-in-career talent that works as well as mid-career talent used to? In other words, during their education, how do they learn to become the boss of agents, such that they are able to command a team, able to produce the same type of work a medium- or large-size team would produce. And we expect that early-in-career folks will be able to do that work now with the aid of these tools."

Jared Spataro,
Chief Marketing Officer,
AI at Work

A Changing Job Market

According to LinkedIn's "[AI and the Global Economy: Unlocking Growth and Reshaping Work](#)" report, the job market also reflects the need for AI skills.

- By 2030, 70% of the skills used most in jobs will change—with AI as the primary catalyst for the change.
- In 2024, hiring for AI technical talent grew 30% faster than overall talent.
- The percentage of jobs on LinkedIn listing an AI literacy skill increased more than six times in the past year.
- 66% of leaders say they wouldn't hire someone without AI literacy skills.
- In the past year, the number of AI literacy skills added by LinkedIn members increased by 177%.

Introducing AI Fluency to the Curriculum

Educators and education leaders acknowledge the need for more AI skills training, with 54% of global educators and 76% of global leaders viewing AI literacy as an essential component of basic education for every student. In the US, 79% of higher education educators say AI literacy is essential.

At Auburn University, a “teaching with AI” course helps faculty incorporate AI into their curriculum, while formal training programs in AI help students and staff continuously improve their AI skills.

In a collaboration between Microsoft and World Wide Technology, 60 K–12 educators from several metro Atlanta school districts received training in AI literacy this spring, covering everything from prompting techniques to how to use AI to personalize learning for individual students. “As we prepare students for jobs that do not yet exist, integrating AI into education ensures they develop critical problem-solving and computational thinking skills,” one participant said.

At the University of Waterloo, an AI-powered tool helps students search for and land jobs—while the school’s Waterloo Experience Accelerate program helps them prepare for AI-driven careers, with hands-on training in everything from cloud computing to data analytics.

Recommendations

- 1 **Learn more about the impact** of AI at work across all industries to better understand job and skill evolution.
- 2 **Prepare students for success** by incorporating AI and human skills like emotional intelligence and analytic judgment into curriculums.
- 3 **Help students use AI** as a competitive advantage in the job hunt.



4

Reimagining the Future of Education



Reimagining the Future of Education

AI is evolving from being a simple time saver and productivity booster to a 24/7 assistant and force multiplier. It can ready students to tap into their creativity to bring projects to life—and to the world.

But beyond its day-to-day benefits, AI can enable institutions and leaders to completely reimagine education experiences, helping them solve long-standing challenges, engage hard-to-reach students, and encourage experimentation and collaboration.



Bringing Opportunity to Life for Students

At the UK's National Youth Theatre, AI helps young performers sharpen their improv skills by generating scene prompts and on-the-fly plot twists, while simultaneously improving their AI skills through their interactions with the latest creative tools.

At a Belgian school where 70 languages are spoken by 425 students, educators use the Microsoft Teams Reading Progress tool to analyze students' reading abilities in real time. Data on reading speed, mistakes, and pronunciation provides feedback that can significantly accelerate a student's progress.

AI also helps would-be entrepreneurs develop new startup ideas and accelerate their time to market. At Babson College, teams of students work alongside Copilot to generate prototype images and help with market research and income statements.

Fostering an AI-ready Campus

Approaching what they don't know with a growth mindset will be key for education leaders in developing institutional AI strategies.

A new IDC white paper, sponsored by Microsoft, "A Blueprint for AI-Ready Campuses: Strategies from the Frontlines of Higher Education," outlined critical strategies based on successful case studies of four forward-looking colleges in the US. "AI is a strategic capability in our broader toolset to achieve our larger goals," said Patty Patria, CIO of Babson College. Among many organizational, process, and technological recommendations, the white paper encouraged leaders to:

- Ensure that any investment in AI aligns with your institution's ultimate vision and goals
- Foster an environment of interdisciplinary collaboration between academic and IT leaders across disciplines
- Promote working groups across faculty, staff, and students to leverage expertise—and to train and learn together

A Thought Partner

Asking AI for assistance with quick answers is one thing; but the real opportunity lies in helping students better understand how to get there on their own. In a Microsoft Research study, researchers found that learners gain a deeper comprehension of course material when given LLM-based explanations versus just the correct answers. Exposure to those explanations also increased how much students felt they learned—and decreased how difficult the problems seemed to them.

“How do you structure writing assignments so that you’re helping kids build the skills that you want them to build, and maybe also integrating the new tools and ways of building those skills we have because of AI? The student articulates an argument, outlines a draft of that argument, and then critiques, refines, and polishes it with AI as a thought partner.”

Jake Hofman,
Senior Principal Researcher
at Microsoft Research

Exploring New Opportunities with AI

Brisbane Catholic Education (BCE), which encompasses more than 140 schools, brought Microsoft 365 Copilot to 12,500 of its educators and support staff, resulting in a reduced administrative load and more time to support students—essential for a profession with high burnout and attrition risk. BCE also provided students aged 13 and up access to Microsoft 365 Copilot Chat to aid in brainstorming and confidence building, resulting in a 275% increase in learner agency for at-risk cohorts.

In one study conducted in a master's program at Indiana University's Kelley School of Business, students performed two tasks—one group using Microsoft 365 Copilot, the other with no AI assistance. The group using Copilot significantly improved their performance, increasing grades by 10% and reducing the time taken for the task by 40%. However, the use of Copilot diminished students' perceptions that the work was their own—highlighting a novel tension between learning efficiency and the intrinsic value of learning.

Critical educational data is often unstructured, and British Columbia's Coquitlam School District transformed that data into actionable insights to make more informed real-time decisions. "Intelligent, AI-supported tools like Copilot will give us the ability to gain insights that will change the educational world," said Assistant Superintendent and CIO Stephen Whiffin.

Sparkling Curiosity and New Ideas

In a survey of university students and educators in the UK, students looked to AI to explain study materials to them, summarize and search for information, and to help spark ideas in collaborative give-and-take sessions ("what would you suggest reading further?," one asked his AI "tutor"). They reported that AI helped them overcome creative blocks and learn beyond the school-provided material, and appreciated that AI-driven assistance was available around the clock.

"When kids are curious, motivation isn't an issue. When we look at effective learning design using technology, it has to come from a place of: how can we leverage AI to spark curiosity? It's not just a pathway to employability, but it's a pathway to be able to fully participate in civic society, to fully participate in the social fabric of the future world. It's those very human qualities that are going to make the difference between those who are successful and those who might be left behind."

Mark Sparvell,
Director, Marketing Education,
Microsoft

Recommendations

- 1 **Align AI efforts to institutional goals**
as a way of accelerating progress.
- 2 **Think big about the opportunity to reimagine**
education experiences and processes.
- 3 **Design opportunities to deepen learning**
and spark collaboration and curiosity with AI.

